

Rafi Khan

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Education

The Ohio State University

M.S. Mechanical Engineering

Jan 2025 – Dec 2026

B.S. Mechanical Engineering

GPA: 3.68/4.0 – Dec 2024

Experience

Graduate Research Assistant, Materials at Extremes Lab, OSU

Jan 2025 – Present

- Research projects focused on computational fatigue and fracture using nCodeDesignLife

SpaceX, Hawthorne, CA

May 2023 – August 2023

Supply Chain Reliability Intern, Starlink Aviation

- Implemented high-voltage testing of Starlink harnesses for the first time in the department. Owned all the processes from test hardware setup, fabrication, programming, and its implementation in the reliability lab.
- Uncovered novel defects when performing tests per Federal Aviation Administration (FAA) requirements
- Led the assessment of the technical qualifications of a Starlink Aviation supplier. Identified gaps in the production process by analyzing manufacturing techniques and corrected the tensile-strength tests under IPC-620 standards (Requirements and Acceptance for Cable and Wire Harness Assemblies)
- Identified a non-conformance (NC) on over 3000 in-stock harnesses planned for installation on Airbus A320s, placing NCs into containment and passing FAA certification

General Electric Aerospace, Dayton, OH

September 2022 – December 2022

Manufacturing Engineering Intern, Aviation Systems

- Hands-on experience working in the New Product Integration Lab for the Hybrid Aircraft high-voltage (1500 V) power converters
- Reviewed drawings and design requirements to assemble complex electrical units and created work instructions for future assemblies
- Designed and fabricated a jig to screen print sensitive semiconductor modules with thermal paste. Leveraged prior work experience at Denso to ensure the process was repeatable
- Worked alongside engineers and technicians and was mentored in fabricating fiber optic cables, aircraft harnesses, and IPC-610 Standard Soldering

Denso, Maryville, TN

May 2022 – August 2022

Process Design Intern

- Responsible for optimizing a high-volume automated production line of computer modules used in 90% of General Motor vehicles
- Designed a calibration box using SolidWorks to reduce downtime in final packaging and save on average \$2000 per production day
- Participated in Kaizen activities and conducted process capability analysis studies (Cpk) to ensure high repeatability

Project

Personal: Thrust Vector Controlled Model Rocket

<https://rafikhan3.github.io>

- Designed a Thrust Vectoring Control (TVC) mount in SolidWorks with 2-axis gimbaling to control a rocket's trajectory during flight
- Used an IMU (Inertial Measurement Unit) to obtain the rocket's orientation and used Simulink to tune a PID controller for a stabilized flight

Skills

Ansys, nCode DesignLife, SolidWorks, HSMWorks (CAM), LabVIEW, MATLAB, Simulink, Java, 3D-Printing, Soldering